Simulation and Experimental Validation of Single-Phase Immersion Cooling System Combined with Heat Pipes

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Introduction

Driven by AI's surging computing demand, data center cooling is projected to consume 40% of total electricity by 2030, resulting in a high PUE of around 1.6. Traditional single-phase immersion cooling (SPIC) relies on power-consuming external water pumps, which causes localized heat accumulation at low flow rates. To address this, we propose a low-energy, pump-free SPIC design combining finned heat pipes and an internal circulation fan. The internal fan effectively breaks the thermal boundary layer, enabling the heat pipes to dissipate a 100W heat load through phase-change heat transfer. By completely eliminating external pumps, this design minimizes non-computing energy consumption, aiming to achieve an optimal PUE below 1.3.

Problem Statement

Traditional single-phase immersion cooling (SPIC) systems heavily rely on external water pumps for fluid circulation, which significantly increases non-computing energy consumption and overall PUE. However, simply eliminating the external pump to save energy creates a new challenge: the lack of external forced convection causes severe localized heat accumulation on the surface of working components. At low flow rates, natural convection is insufficient, leading to the formation of a thermal boundary layer around the high-power chips. Therefore, the core problem is how to effectively implement internal flow disruption to break the thermal boundary layer and prevent temperature surging, entirely without the energy penalty of external pumps.

Design and Methods

This study integrates practical prototype experiments and numerical simulations to evaluate the pump-free cooling system.

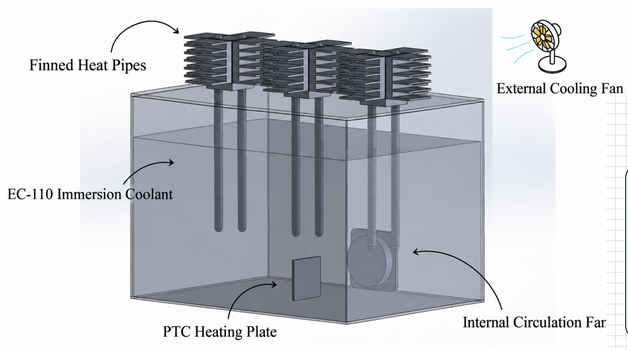


Fig. 1. Experimental setup of the pump-free cooling system.

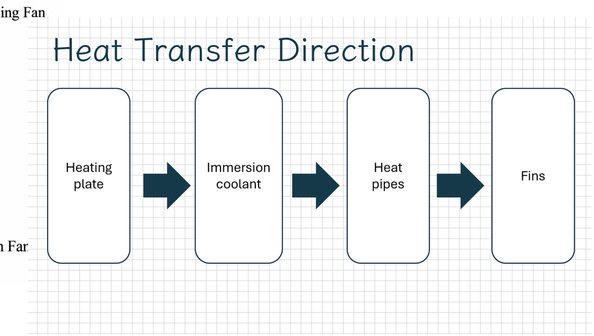


Fig. 2. Heat transfer path diagram.

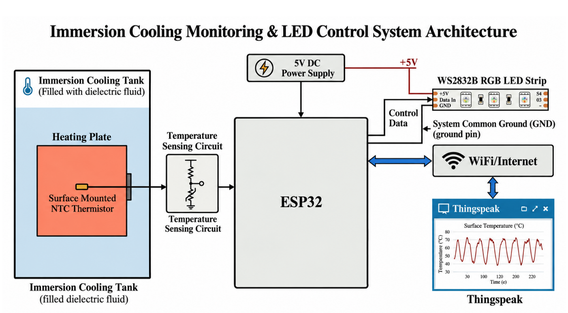


Fig. 3. Temperature measurement system.



Fig. 4. Detect the temperature and PUE in time.

Simulation and Results

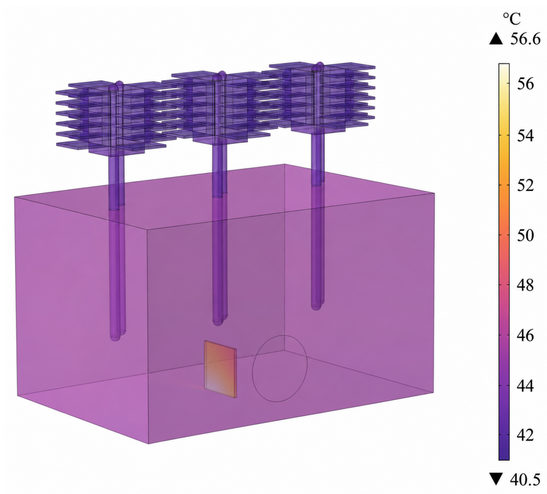


Fig. 5. COMSOL Simulation result. It demonstrates the temperature distribution in this system and the maximum temperature is lower than the limit 85°C.

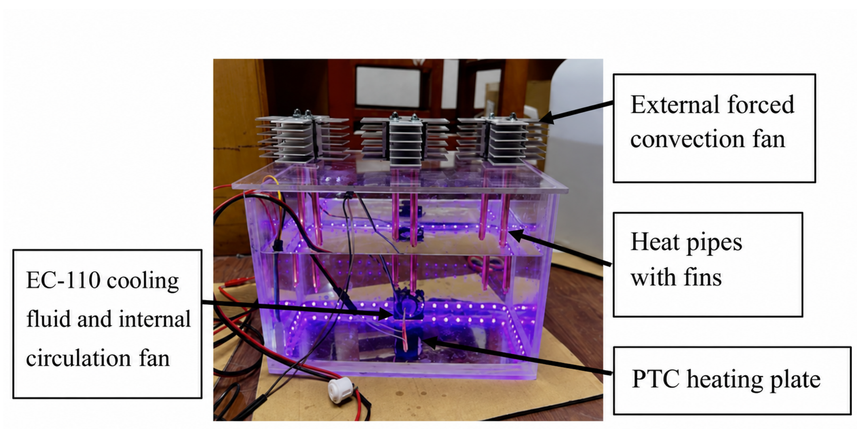


Fig. 6. Practical model. The prototype of the pump-free immersion cooling system, featuring finned heat pipes, an internal circulation fan, and a PTC heating plate.

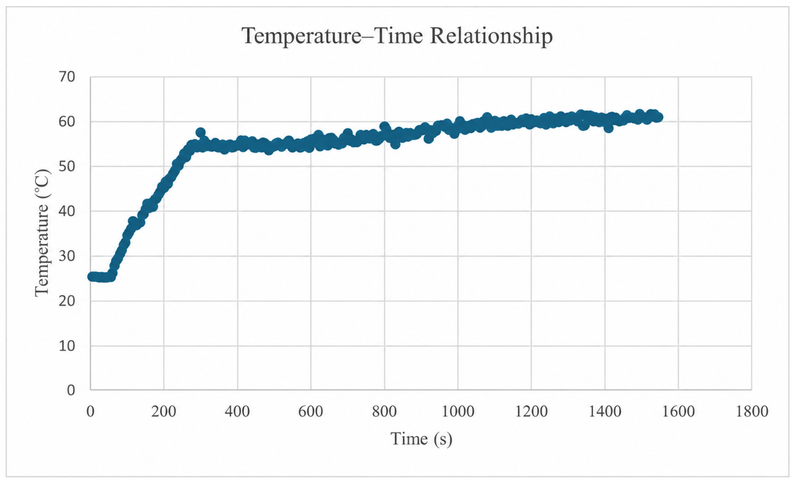


Fig. 7. Temperature-Time relationship of the heating plate. It demonstrates the system reaching thermal equilibrium at approximately 60°C.

Conclusion

This study successfully demonstrates the feasibility of a pump-free single-phase immersion cooling (SPIC) system. By integrating an internal circulation fan and finned heat pipes, the system effectively breaks the thermal boundary layer and dissipates a 100W heat load without relying on external water pumps. Crucially, cross-verification between the practical prototype and COMSOL simulations yielded a steady-state error of less than 5%. This confirms that the numerical model accurately predicts the fluid and thermal field behaviors. Furthermore, this highly validated simulation model allows us to easily test different operating scenarios and optimize the system design in the future without incurring significant physical prototyping costs. Ultimately, this research establishes a reliable foundation for achieving an optimal PUE below 1.3.

Acknowledgement

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Demo Video

Scan for demo video

References:

[1] Huang, Y. P., Ge, J. L., Chen, Y. P., & Zhang, C. B. (2023). Natural and forced convection heat transfer characteristics of single-phase immersion cooling systems for data centers.

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